

Class Field Theory

1 0908

Goal Describe all abelian extension of number fields.
i.e. inverse Galois problem for abelian groups.

Recall (Kronecker-Weber Theorem) Every abelian extension of the rational numbers is contained in a cyclotomic field.

What happens if we replace $\mathbb{Q}$ by a number field K ?

Denote $\text{Cl}(K)$ as the ideal class group of K . (Fractional ideals modulo principal ideals.)

Theorem 1.1 : Let L be the maximal unramified abelian extension of a number field K . Then L/K is finite and $\text{Gal}(L/K) \cong \text{Cl}(K)$.
 L is called the Hilbert class field of K .

Recall Let $N > 1$ be an integer. ζ_N denotes a primitive N -th root of unity. Then

1. $\text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N\mathbb{Z})^\times$.
2. Suppose $N \not\equiv 2 \pmod{4}$. Then p is ramified in $\mathbb{Q}(\zeta_N)$ if and only if $p \mid N$.
3. If p splits completely in $\mathbb{Q}(\zeta_N)$, then p is unramified, so the decomposition group is isomorphic to $\text{Gal}(\kappa(P)/\kappa(p))$.

The Frobenius element $\text{Frob}(P/p)$ is the unique element in the decomposition group that acts as $x \mapsto x^p$ on the residue field.

It is too restrictive to ask for unramified extensions.

So we ask for abelian extensions.

1.1 Review $\mathbb{A}_K, \mathbb{I}_K = \mathbb{A}_K^\times, \text{Cl}(K)$

Let I be a fractional ideal in K . Then I can be written as $I = P_1^{a_1} \dots P_r^{a_r}$, where P_i are prime ideals and $a_i \in \mathbb{Z}$.

So the group of fractional ideals is a free abelian group generated by the prime ideals.

$$\bigoplus_P \mathbb{Z}$$

and Cl is the quotient of $\bigoplus_P \mathbb{Z}$ by the image of the map $K^\times \rightarrow \bigoplus_P \mathbb{Z}$ given by $x \mapsto (v_P(x))_P$, where v_P is the valuation at P .

Let $U = \prod_{v:\text{inf}} K_v^\times \times \prod_{v:\text{fin}} \mathcal{O}_v^\times$. It is an open subgroup of $\mathbb{A}_K^\times$.

Then $\mathbb{A}_K^\times/U$ is exactly $\bigoplus_{v:\text{fin}} \mathbb{Z}$. So we have a surjective map $\mathbb{A}_K^\times \rightarrow \text{Cl}(K)$, and the kernel is $UK^\times$.

As the class group corresponds to the maximal unramified abelian extension, we want to shrink U to a smaller open subgroup to detect more abelian extensions.

[Ray class group]@Obsidian

Remark: This isomorphism shows that $\text{Cl}(K, \mathfrak{a})$ is finite:

By the isomorphism, it is discrete.

Also, it is a quotient of $\mathbb{A}_K^1/K^\times$, which is compact, so it is finite.

Example 1.2 : $\text{Cl}(\mathbb{Q}, N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$.

[計算 $\text{Cl}(\mathbb{Q}, N)$]@Obsidian

By Kronecker-Weber, we know that $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ is the inverse limit of $(\mathbb{Z}/N\mathbb{Z})^\times$ over N .

By above, $(\mathbb{Z}/N\mathbb{Z})^\times$ is a quotient of $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times$.

Class field theory is saying that we can construct a map from $C_K = \mathbb{A}_K^\times/K^\times$ to $\text{Gal}(K^{\text{ab}}/K)$ for any number field K , and the image of this map is dense.

Theorem 1.3 (Local class field theory): Let K be a local field. There is a unique continuous homomorphism

$$\rho_K : K^\times \rightarrow \text{Gal}(K^{\text{ab}}/K)$$

satisfying the following properties:

1. For any L/K finite abelian extension, the kernel of ρ_K is exactly $N_{L/K}(L^\times)$, the norm of $L^\times$ in $K^\times$.

$$\begin{array}{ccccc} K^\times & \longrightarrow & \text{Gal}(K^{\text{ab}}/K) & \twoheadrightarrow & \text{Gal}(K^{\text{un}}/K) \\ \downarrow & & \downarrow & \swarrow & \\ \mathbb{Z} & \longrightarrow & \text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) & & \end{array}$$

- 2.
3. There is a 1-1 correspondence between U : an open subgroup of $\text{Gal}(K^{\text{ab}}/K)$ and V : open subgroup of finite index of $K^\times$ given by $U \mapsto \rho_K^{-1}(U)$.

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Definition (CSA): A (finite-dimensional) central simple algebra (CSA) over a field K is a finite-dimensional K -algebra R such that:

- R is a finite-dimensional K -algebra.
- R is central: The center of R is exactly K .
- R is simple: R has no nontrivial two-sided ideals.

Example 2.1 : The matrix algebra $M_{n(K)}$ is a CSA over K .

Fact 2.2 (Wedderburn): Let A be a CSA over K . Then there's a central division algebra D s.t.

$$A \cong M_n(D)$$

for some $n \in \mathbb{Z}_{>0}$ and D is unique up to isomorphism.

Definition Let A, B be two CSAs over K . We say A and B are Brauer equivalent if there exist $m, n \in \mathbb{Z}_{>0}$ s.t.

$$A \otimes_K M_m(K) \cong B \otimes_K M_n(K)$$

i.e. their corresponding division algebras are isomorphic. The Brauer equivalence class of A is denoted by $[A]$.

The Brauer group $\text{Br}(K)$ is the set of Brauer equivalence classes of CSAs over K with the group operation given by tensor product over K .

Then $[A]^{-1} = [A^{\text{op}}]$.

The importance of the Brauer group in CFT:

If K_v is non-archimedean, then $\text{Br}(K_v) \cong \mathbb{Q}/\mathbb{Z}$ and it is canonical.

We can write $\text{Br}(K_v)$ explicitly as its element is the form (χ, a) where χ is a character of $\text{Gal}(K_v^{\text{ab}}/K_v)$ and $a \in K_v^\times$.

It help us define a map

$$\begin{aligned} \rho : K_v^\times &\rightarrow \text{Hom}(\text{Hom}(\text{Gal}(K_v^{\text{ab}}/K_v), \mathbb{Q}/\mathbb{Z}), \mathbb{Q}/\mathbb{Z}) \\ a &\mapsto (\chi \mapsto \text{inv}(a, \chi)) \end{aligned}$$

and the right hand side is $\text{Gal}(K_v^{\text{ab}}/K_v)$ given by Pontryagin duality.

3 Lecture 1: Group cohomology

Let G be a finite group.

Definition

1. M is a G -module if M is an abelian group with a G -action that is compatible with the group structure.

We also consider M as a $\mathbb{Z}[G]$ -module.

2. $f : M \rightarrow N$ is a G -module homomorphism if f is a map preserving the G -action.

Definition $C^n(\mathbb{Z}[G], A) = \text{Hom}_{G(\mathbb{Z}[G^{n+1}], A)}$.

$\partial^n : C^n(\mathbb{Z}[G], A) \rightarrow C^{n+1}(\mathbb{Z}[G], A)$ is defined by

$$(\partial^n f)(g_0, \dots, g_{n+1}) = \sum_{i=0}^{n+1} (-1)^i f(g_0, \dots, \hat{g}_i, \dots, g_{n+1})$$

It defines a cochain complex $(C^n(\mathbb{Z}[G], A), \partial^n)$, and the cohomology of this complex is denoted by $H^n(G, A)$.

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Definition A G -module B is called relatively injective if B is a direct summand of an induced module $\text{Ind}_1^G(A)$ for some abelian group A .

Corollary 4.1 : A relatively injective G -module is acyclic.

4.1 Change of groups

General: $f : G' \rightarrow G$: group homomorphism of finite groups.

Let A be a G -module. Through f , we can make A a G' -module. Denote as f^*A . This is a functor $f^* : G\text{-Mod} \rightarrow G'\text{-Mod}$. We have a natural map $f^* : H^n(G, A) \rightarrow H^n(G', f^*A)$ for all n .

Now consider the case $H \hookrightarrow G$. The pullback map is called the restriction map.

Theorem 4.2 (Shapiro lemma):

$$H^n(H, A) \cong H^n(G, \text{Ind}_H^G(A))$$

Proof: By Frobenius reciprocity, we have

$$\text{Hom}_G(B, \text{Ind}_H^G(A)) \cong \text{Hom}_H(B, A)$$

For $n = 0$: Take $B = \mathbb{Z}$ with trivial action.

Now for $n > 0$, take a injective resolution $A \rightarrow I^\bullet$. By Frobenius reciprocity and the exactness of induction functor, $\text{Ind}_H^G(I^\bullet)$ is an injective resolution of $\text{Ind}_H^G(A)$.

Applying the case $n = 0$ to $I^\bullet$, we get the desired isomorphism since both sides are universal δ -functors.

■

Consider another case $H \hookrightarrow G$ and A is a G -module. Consider the following map:

For $n = 0$, we have a map $H^0(H, A|_H) \rightarrow H^0(G, A)$ given by $a \mapsto \sum_{g \in G/H} ga$.

Now, $H^n(H, A|_H)$ is a δ -functor. We want to show that it's the derived functor of $A \mapsto A^H = H^0(H, A|_H)$.

By the dual of Frobenius reciprocity,

$$\mathrm{Hom}_H(- \otimes_H G, I) \cong \mathrm{Hom}_G(-, I)$$

Since $- \otimes_H G$ is exact, $I^\bullet|_H$ is an injective resolution of $A|_H$. (Restriction is exact.)

This proves that $H^n(H, A|_H)$ is the derived functor of $A \mapsto A^H$ hence a universal δ -functor.

As $A \mapsto H^n(G, A)$ is also a δ -functor, we have a natural transformation $H^n(H, A|_H) \rightarrow H^n(G, A)$ for all n .

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5.1 Inflation

Let $H \triangleleft G$ be a normal subgroup.

Given a G -module A , we can make A^H a G/H -module.

Then we have a natural map

$$\mathrm{inf}_{G/H} : H^n(G/H, A^H) \rightarrow H^n(G, A^H) \rightarrow H^n(G, A)$$

Theorem 5.1 : We have an exact sequence

$$0 \longrightarrow H^1(G/H, A^H) \xrightarrow{\mathrm{inf}_{G/H}} H^1(G, A^H) \xrightarrow{\mathrm{res}} H^1(H, A)$$

5.2 Intermezzo: Spectral sequences (arising from a double complex)

Set-up: $M^{\bullet, \bullet}$: bounded below double complex of abelian groups.

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M^{0,0} & \longrightarrow & M^{0,1} & \longrightarrow & M^{0,2} \longrightarrow \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M^{1,0} & \longrightarrow & M^{1,1} & \longrightarrow & M^{1,2} \longrightarrow \\ & & \downarrow & & \downarrow & & \downarrow \end{array}$$

with

- $d_{\rightarrow} \circ d_{\rightarrow} = 0$
- $d_{\downarrow} \circ d_{\downarrow} = 0$
- $d_{\rightarrow} \circ d_{\downarrow} + d_{\downarrow} \circ d_{\rightarrow} = 0$

Define the total complex $M^\bullet$ by $M^n = \sum_{p+q=n} M^{p,q}$ and $d = d_{\rightarrow} + d_{\downarrow}$.

We want to compute the cohomology of $M^\bullet$.

The strategy is to filter $M^\bullet$ by p and q and get two spectral sequences.

Define $F^i M^n = \sum_{p \geq i} M^{p, n-p}$.

Then

$$F^i H^n = (Z^n \cap F^i M^n) + B^n / B^n \subset H^n$$

It gives a filtration of H^n :

$$0 \subset F^n H^n \subset F^{n-1} H^n \subset \dots \subset F^0 H^n = H^n$$

- Page 0: $E_0^{p,q} = M^{p,q}$
- Page 1: $E_1^{p,q} = H^q(M^{p,\bullet})$ with the differential $d_1^{p,q} : E_1^{p,q} \rightarrow E_1^{p+1,q}$ induced by $d_\downarrow$.
- Page 2: $E_2^{p,q} = H^p(E_1^{\bullet,q})$ with the differential $d_2^{p,q} : E_2^{p,q} \rightarrow E_2^{p+2,q-1}$ induced by $d_\downarrow$:

eye The map is given by:

For $[x] \in E_2^{p,q}$, we have $x \in M^{p,q}$ s.t. $d_\rightarrow x = 0$ and $d_\downarrow x = d_\rightarrow(-y)$, for some $y \in M^{p+1,q-1}$. Then $d_2^{p,q}([x]) = [d_\downarrow^{p+1,q-1}(y)]$.

Inductively, we can define

$$E_{r+1}^{p,q} = \ker(d_r^{p,q}) / \text{im}(d_r^{p-r,q+r-1})$$

and

$$d_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}$$

For a fixed (p, q) , the sequence $E_r^{p,q}$ stabilizes for r large enough. Denote the stable value by $E_\infty^{p,q}$.

If for some $x \in M^{p,q}$, we have $d_\rightarrow x = 0$ and $d_\downarrow x = 0$, i.e. $dx = 0$. Then $d_n(x) = 0$ for all n , so $[x] \in E_\infty^{p,q}$.

Similarly, if $x \in M^{p,q}$, $y \in M^{p+1,q-1}$ and $d(x+y) = 0$, then $[x+y] \in E_\infty^{p,q}$.

In fact,

$$E_\infty^{p,q} = \frac{Z^n \cap F^p M^{p+q}}{Z^n \cap F^{p+1} M^{p+q} + B^n \cap F^p M^{p+q}}$$

which is exactly the p -th graded piece of the filtration of H^n i.e. $\frac{F^p H^n}{F^{p+1} H^n}$.

Now, for lower degree terms, we have

- $F^1 H^1 = E_\infty^{1,0}$
- $H^1 / F^1 H^1 = E_\infty^{0,1}$
- $F^2 H^2 = E_\infty^{2,0}$

i.e.

$$0 \longrightarrow E_\infty^{1,0} \longrightarrow H^1 \longrightarrow E_\infty^{0,1} \longrightarrow 0$$

We have $E_\infty^{1,0} = E_2^{1,0}$ and $E_\infty^{0,1} = \ker(d_2^{0,1})$.

Also, $E_\infty^{2,0} = \text{coker}(d_2^{0,1})$.

So we have an exact sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \longrightarrow H^2$$

Theorem 5.2 (Grothendieck spectral sequence): Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be three abelian categories.

$F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{C}$ be two additive left-exact functors.

Assume that

- $\mathcal{A}, \mathcal{B}$ have enough injectives.
- F sends injective objects of $\mathcal{A}$ to G -acyclic objects of $\mathcal{B}$.

Then

$$E_2^{p,q} = (R^p G)(R^q F)(A) \implies R^{p+q}(G \circ F)(A)$$

For $A \in \mathcal{A}$, take an injective resolution $A \rightarrow I^\bullet$.

Then $F(I^\bullet)$ is a complex in $\mathcal{B}$ of acyclic objects.

Fact 5.3 : There is a double complex $J^{p,q}$ s.t. $J^{p,\bullet}$ is an injective resolution of $F(I^p)$ for all p .

Example 5.4 :

- $\mathcal{A} = \text{Mod}_G, \mathcal{B} = \text{Mod}_{G/H}, \mathcal{C} = \text{Ab}$.
- $F(A) = A^H, G(B) = B^{G/H}$.

Then $R^p G = H^p(G/H, -)$, $R^q F = H^q(H, -)$, and $R^{p+q}(G \circ F) = H^{p+q}(G, -)$.

So we have a spectral sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \longrightarrow H^2$$

which can be rewritten as

$$0 \longrightarrow H^1(G/H, A^H) \xrightarrow{\inf_{G/H}} H^1(G, A) \xrightarrow{\text{res}} H^1(H, A)^{G/H} \xrightarrow{d_2} H^2(G/H, A^H) \longrightarrow H^2(G, A)$$

5.3 Application

Let G be a finite group and A be a G -module.

Theorem 5.5 : If $H \leq G$ is a subgroup of index m , then

$$\text{cor}_G^H \circ \text{res}_G^H = m \text{id}_{H^n(G, A)}$$

Proof: For $n = 0$, it is clear.

By dimension shifting, we are done. ■

Corollary 5.6 : Take $H = 1$, then we know that $H^n(G, A)$ is of $|G|$ -torsion for $n > 0$.

Corollary 5.7 : If A is of finite type, then $H^n(G, A)$ is finite for $n > 0$.

Proof: By the previous corollary, $H^n(G, A)$ is of $|G|$ -torsion.

Also, $H^n(G, A)$ is finitely generated abelian group since A is finitely generated by the cochain complex construction.

So it is finite. ■

Corollary 5.8 : Suppose that A is uniquely divisible i.e. A is a $\mathbb{Q}$ -vector space. Then $H^n(G, A) = 0$ for $n > 0$.

Proof: Multiplication by $|G|$ is an automorphism of A since A is uniquely divisible.

But it is zero map on $H^n(G, A)$ for $n > 0$. So $H^n(G, A) = 0$ for $n > 0$. ■

Example 5.9 : Consider the exact sequence of G -modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

with the trivial G -action. Then by the long exact sequence of cohomology, we have

$$H^1(G, \mathbb{Q}/\mathbb{Z}) \cong H^2(G, \mathbb{Z})$$

Now suppose G is a Galois group of a finite Galois extension L/K .

Since $\mathbb{Q}/\mathbb{Z}$ is of trivial G -action, we have

$$H^1(G, \mathbb{Q}/\mathbb{Z}) \cong \text{Hom}(G, \mathbb{Q}/\mathbb{Z}) = \text{Hom}(G^{\text{ab}}, \mathbb{Q}/\mathbb{Z})$$

As any finite subgroup of $\mathbb{Q}/\mathbb{Z}$ is cyclic, we have

$$G/\ker f \cong \text{im } f \leq \mathbb{Q}/\mathbb{Z}$$

and $\text{im } f \cong \mathbb{Z}/n\mathbb{Z}$ for some n .

So to give a f is equivalent to giving a cyclic quotient of G and a generator of the cyclic group $\text{Gal}(L^{\ker f}/K)$.

Example 5.10 : Same as above. We have $H^1(G, \mathbb{Z}) = 0$.

6 Lecture 2: Tate cohomology

Definition Let G be a finite group.

1. The element norm $\alpha_0 \in \mathbb{Z}[G]$ is defined as

$$\alpha_0 = \sum_{g \in G} g$$

2. Let A be a G -module. The norm map $N : A \rightarrow A$ is defined as

$$N(a) := \alpha_0 a = \sum_{g \in G} ga$$

3. The augmentation (two-sided) ideal I_G is defined as

$$I_G = \ker(\mathbb{Z}[G] \rightarrow \mathbb{Z}) = \text{span}_{\mathbb{Z}}(g - 1 : g \in G)$$

4. $A_G := A/I_G A = \mathbb{Z}[G]/I_G \otimes_{\mathbb{Z}[G]} A$, which is the maximal quotient of A on which G acts trivially.

As the functor $A \mapsto A_G$ is given by a tensor product, it is right-exact.

So we can define its left derived functor, denote by $H_n(G, A)$.

Example 6.1 : $H_1(G, \mathbb{Z}) = G^{\text{ab}}$.

Proof: Consider the exact sequence of G -modules

$$0 \longrightarrow I_G \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0$$

We have the long exact sequence of homology

$$0 \longrightarrow H_1(G, \mathbb{Z}) \longrightarrow I_G/I_G^2 \longrightarrow \mathbb{Z}[G]_G = \mathbb{Z} \longrightarrow \mathbb{Z}_G = \mathbb{Z} \longrightarrow 0$$

So $H_1(G, \mathbb{Z}) \cong I_G/I_G^2$.

Consider the map $G \rightarrow I_G/I_G^2$ given by $g \rightarrow [g - 1]$.

Since $I_G = \text{span}_{\mathbb{Z}}(g - 1 : g \in G)$, the map is surjective.

Also, since I_G/I_G^2 is abelian, the map factors through G^{ab} .

Now consider the map $I_G \rightarrow G^{\text{ab}}$ given by $g - 1 \rightarrow g$.

It is surjective and factors through I_G/I_G^2 .

6.1 0929

Definition (Tate cohomology): Let G be a finite group and A be a G -module. The Tate cohomology groups $\hat{H}^n(G, A)$ are defined as follows:

- For $n > 0$, $\hat{H}^n(G, A) = H^n(G, A)$.
- For $n = 0$, $\hat{H}^0(G, A) = A^G / N_G(A)$.
- For $n = -1$, $\hat{H}^{-1}(G, A) = \ker(N_G) / I_G A$.
- For $n < -1$, $\hat{H}^n(G, A) = H_{-n-1}(G, A)$.

where N_G is the norm map and I_G is the augmentation ideal.

Theorem 6.2 : Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be a short exact sequence of G -modules. Then we have a long exact sequence of Tate cohomology groups

$$\begin{array}{ccccccc}
 \dots & \xrightarrow{\delta} & \hat{H}^{-2}(G, A) & \longrightarrow & \hat{H}^{-2}(G, B) & \longrightarrow & \hat{H}^{-2}(G, C) \\
 & & & & \searrow \delta & & \\
 & & \hat{H}^{-1}(G, A) & \longrightarrow & \hat{H}^{-1}(G, B) & \longrightarrow & \hat{H}^{-1}(G, C) \\
 & & & & \searrow \delta & & \\
 & & \hat{H}^0(G, A) & \longrightarrow & \hat{H}^0(G, B) & \longrightarrow & \hat{H}^0(G, C) \\
 & & & & \searrow \delta & & \\
 & & \hat{H}^1(G, A) & \longrightarrow & \hat{H}^1(G, B) & \longrightarrow & \hat{H}^1(G, C) \xrightarrow{\delta} \dots
 \end{array}$$

Proof: We have the following exact sequences of abelian groups:

$$\begin{array}{ccccccccc}
 H_1(G, C) & \xrightarrow{\delta} & A_G & \longrightarrow & B_G & \longrightarrow & C_G & \longrightarrow & 0 \\
 & & \downarrow N & & \downarrow N & & \downarrow N & & \\
 0 & \longrightarrow & A^G & \longrightarrow & B^G & \longrightarrow & C^G & \xrightarrow{\delta} & H^1(G, A)
 \end{array}$$

by the snake lemma, we get the desired long exact sequence of Tate cohomology groups. ■

Similar to the case of group cohomology, we have the following results:

The group cohomology can be computed by applying the functor $\text{Hom}_{G(-, A)}$ to the exact sequence

$$\dots \longrightarrow \mathbb{Z}[G^2] \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0$$

and compute the cohomology of the resulting complex.

For the group homology, we can apply the functor $- \otimes_{\mathbb{Z}[G]} A$ to the same exact sequence and compute the homology of the resulting complex.

But for G finite, we have an isomorphism between $\mathbb{Z}[G^n] \otimes_{\mathbb{Z}[G]} A$ and $\text{Hom}_{G(\mathbb{Z}[G^n], A)}$. So we can compute the Tate cohomology by applying the functor $\text{Hom}_{G(-, A)}$ to the exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z}[G^2] \longrightarrow \dots$$

where the boundary map is given by

$$\partial_n(g_1, \dots, g_n) = \sum_{g \in G} \sum_{i=0}^n (-1)^i (g_1, \dots, g_{i-1}, g, g_i, \dots, g_n)$$

combining the two complexes, it is

$$\begin{array}{ccccccc} \dots & \longrightarrow & \mathbb{Z}[G^2] & \longrightarrow & \mathbb{Z}[G] & & \\ & & & \searrow & & & \\ & & \mathbb{Z}[G] & \longrightarrow & \mathbb{Z}[G^2] & \longrightarrow & \dots \end{array}$$

and the map $\mathbb{Z}[G] \rightarrow \mathbb{Z}[G]$ is given by the composition $\mathbb{Z}[G] \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}[G]$, which is exactly the norm map.

We can similarly define restriction and corestriction maps for Tate cohomology.

We first define them for homology. Let $H \leq G$ be a subgroup of finite index.

Note that both $H_n(G, A)$ and $H_n(H, A|_H)$ are universal δ -functors. So we only need to define the map for $n = 0$. (need to check restriction map $A \rightarrow A|_H$ preserves projectivity and exactness)

- Corestriction: $A_H \rightarrow A_G$ is given by $a + I_H A \mapsto a + I_G A$.
- Restriction: $A_G \rightarrow A_H$ is given by $a + I_G A \mapsto \sum_{g \in H \backslash G} ga + I_H A$. (check well-defined)

For Tate cohomology, we only need to show that the restriction and corestriction maps are compatible with the norm map, and the result comes from the snake lemma.

We can check it by direct computation.

Theorem 6.3 : If I is a relatively injective G -module, then $\hat{H}^n(G, I) = 0$ for all $n \in \mathbb{Z}$.

Proof: Already know for $n > 0$. For $n \leftarrow 1$, since for G finite, induced modules are isomorphic to coinduced modules, we have I is also relatively projective. So $\hat{H}^n(G, I) = 0$ for $n \leftarrow 1$.

For $n = 0, -1$, it can be checked by direct computation for induced modules. ■

Theorem 6.4 (restriction-corestriction): Let $H \leq G$ be a subgroup of finite index. Then

- $\text{cor}_G^H \circ \text{res}_G^H = [G : H] \text{id}_{\hat{H}^n(G, A)}$
- $\hat{H}^n(G, A)$ is of $|G|$ -torsion.
- If A is of finite type, then $\hat{H}^n(G, A)$ is finite.

Now, we know that $H_1(G, \mathbb{Z})$ (or $\hat{H}^{-2}(G, \mathbb{Z})$) is the abelianization.

What is the restriction map $\text{res}_G^H : H_1(G, \mathbb{Z}) \rightarrow H_1(H, \mathbb{Z})$?

This problem is important because the abelianization of the Galois group is related to the maximal abelian

subextension.

This map is called the transfer map.

$$\begin{array}{ccccccc}
0 & \longrightarrow & H_1(G, \mathbb{Z}) & \longrightarrow & H_0(G, I_G) & \longrightarrow & 0 \\
& & \downarrow \text{res} & & \downarrow \text{res} & & \\
0 & \longrightarrow & H_1(H, \mathbb{Z}) & \longrightarrow & H_0(H, I_G) & \longrightarrow & H_0(H, \mathbb{Z}[G]) \\
& & \uparrow & & \uparrow & & \\
0 & \longrightarrow & H_1(H, \mathbb{Z}) & \longrightarrow & H_0(H, I_H) & \longrightarrow & 0
\end{array}$$

So we know that the transfer map is given by $H_0(G, I_G) \rightarrow H_0(H, I_H)$, which is given by $H_0(G, I_G) \rightarrow H_0(H, I_G)$ and restrict on the image of the injection $H_0(H, I_H) \rightarrow H_0(H, I_G)$.

As $H_0(H, I_G) = I_G/I_H I_G$ and the map $H_0(H, I_H) \rightarrow H_0(H, I_G)$ is given by $a + I_H^2 \mapsto a + I_H I_G$, we know that

$$I_H/I_H^2 \xrightarrow{\sim} (I_H + I_H I_G)/I_H I_G$$

Also, the restriction map $H_0(G, I_G) \rightarrow H_0(H, I_G)$ is the norm map

$$\begin{aligned}
N_{H \setminus G} : I_G/I_G^2 &\rightarrow I_G/I_H I_G \\
x + I_G^2 &\mapsto \sum_{g \in H \setminus G} gx + I_H I_G
\end{aligned}$$

Now, for $x \in G/[G, G]$, it corresponds to $x - 1 \in I_G/I_G^2$.

$$N_{H \setminus G}(x - 1 + I_G^2) = \sum_{g \in H \setminus G} g(x - 1) + I_H I_G = \sum_{g \in H \setminus G} (gx - g) + I_H I_G$$

Take R be a set of representatives of $H \setminus G$.

For each $r \in R$, there is a unique $h_r \in H$ and $r' \in R$ s.t. $rx = h_r r'$.

Moreover, $r \mapsto r'$ is a permutation of R . So we have

$$\begin{aligned}
&= \sum_{r \in R} (h_r r' - r) + I_H I_G \\
&= \sum_{r \in R} (h_r r' - r') + I_H I_G \\
&= \sum_{r \in R} (h_r - 1) + I_H I_G
\end{aligned}$$

which corresponds to $\prod_{r \in R} h_r \in H/[H, H]$.

So we have the following theorem:

Theorem 6.5 : The restriction map $\text{res}_G^H : H_1(G, \mathbb{Z}) \rightarrow H_1(H, \mathbb{Z})$ is given by

$$x \in G/[G, G] \mapsto \prod_{r \in R} h_r \in H/[H, H]$$

where R is a set of representatives of $H \setminus G$ and for each $r \in R$, there is a unique $h_r \in H$ and $r' \in R$ s.t. $rx = h_r r'$.

Definition Let $A \times B \rightarrow C$ be a pairing of G -modules. It induces a pairing

$$C^i(G, A) \times C^j(G, B) \rightarrow C^{i+j}(G, C)$$

given by

$$(f \smile g)(g_0, \dots, g_{i+j}) = \langle f(g_0, \dots, g_i), g(g_i, \dots, g_{i+j}) \rangle$$

Proposition 6.6 : Let A, B, C be 3 G -modules.

1. Suppose

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

and

$$0 \longrightarrow C' \longrightarrow C \longrightarrow C'' \longrightarrow 0$$

be exact sequences of G -modules.

Suppose that we have pairings $A \times B \rightarrow C$, $A' \times B \rightarrow C'$ and $A'' \times B \rightarrow C''$ which are compatible with the exact sequences.

Then

$$(\delta a'') \smile b = \delta(a'' \smile b)$$

where a'' is a cocycle in A'' and b is a cocycle in B .

2. Suppose

$$0 \longrightarrow B' \longrightarrow B \longrightarrow B'' \longrightarrow 0$$

and

$$0 \longrightarrow C' \longrightarrow C \longrightarrow C'' \longrightarrow 0$$

be exact sequences of G -modules.

Suppose that we have three pairings $A \times B \rightarrow C$, $A \times B' \rightarrow C'$ and $A \times B'' \rightarrow C''$ which are compatible with the exact sequences.

Then

$$a \smile (\delta b'') = (-1)^p \delta(a \smile b'')$$

where p is the degree of a .

Proof: Use Leibniz rule and chase the diagram of the snake lemma:

Let α be a lifting of a'' . Then $\delta a'' = [d\alpha]$ and $\delta(a'' \smile b) = [d(\alpha \smile b)]$.

Use Leibniz rule, we have $d(\alpha \smile b) = d\alpha \smile b + (-1)^p \alpha \smile db = d\alpha \smile b$.

For the second part, let β be a lifting of b'' . Then $\delta b'' = [d\beta]$ and $\delta(a \smile b'') = [d(a \smile \beta)]$.

Use Leibniz rule, we have $d(a \smile \beta) = da \smile \beta + (-1)^p a \smile d\beta = (-1)^p a \smile d\beta$.

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Theorem 6.7 : Let G be a finite group. Then there is a unique family of pairings

$$\hat{H}^p(G, A) \times \hat{H}^q(G, B) \rightarrow \hat{H}^{p+q}(G, C)$$

associate to all $p, q \in \mathbb{Z}$ and any pairing $A \times B \rightarrow C$ of G -modules, satisfying:

- $p = q = 0$, the map are induced by the pairing $A^G \times B^G \rightarrow C^G$ given by the restriction of the pairing $A \times B \rightarrow C$.
- These maps are functorial in the category of G -modules.
- These maps are compatible with connecting homomorphisms in the sense of the previous proposition.

Proof: We want to define the pairing on the level of cochains.

We know that the Tate cohomology can be computed by the cochain complex

$$\begin{array}{ccccccc} X_\bullet : \dots & \longrightarrow & \mathbb{Z}[G^2] & \longrightarrow & \mathbb{Z}[G] & & \\ & & & \searrow & & & \\ & & \mathbb{Z}[G] & \longrightarrow & \mathbb{Z}[G^2] & \longrightarrow & \dots \end{array}$$

apply the functor $\text{Hom}_{G(-,A)}$ to the complex, we get a cochain complex computing $\hat{H}^n(G, A)$.

We want to construct $\varphi_{p,q} : X_{p+q} \rightarrow X_p \otimes_{\mathbb{Z}} X_q$ satisfying the following conditions:

- Leibniz rule: $\varphi_{p,q} \circ \partial_{p+q+1} = (\partial_{p+1} \otimes \text{id}) \circ \varphi_{p+1,q} + (-1)^p (\text{id} \otimes \partial_q) \circ \varphi_{p,q+1}$
- For $p = q = 0$, $\pi : \mathbb{Z}[G] \rightarrow \mathbb{Z}$, $(\pi \otimes \pi) \circ \varphi_{0,0} = \pi$.

Assume that we have constructed all $\varphi_{p,q}$, then we can complete the proof by the similar argument as the previous proposition.

We can construct $\varphi_{p,q}$ like this:

- $p, q \geq 0$:

$$\varphi_{p,q}(g_0, \dots, g_{p+q}) = (g_0, \dots, g_p) \otimes (g_p, \dots, g_{p+q})$$

- $p, q \geq 1$:

$$\varphi_{-p,-q}(g_1, \dots, g_{p+q}) = (g_1, \dots, g_p) \otimes (g_{p+1}, \dots, g_{p+q})$$

- $p \geq 0, q \geq 1$:

$$\varphi_{p,-p-q}(g_1, \dots, g_q) = \sum_{(h_1, \dots, h_p) \in G^p} (g_1, h_1, \dots, h_p) \otimes (h_p, \dots, h_1, g_1, \dots, g_q)$$

Other cases are defined in the textbook. It is a long but straightforward computation to check the Leibniz rule and the condition for $p = q = 0$.

Proposition 6.8 :

1. If we identify $(A \otimes B) \otimes C$ with $A \otimes (B \otimes C)$, then for any $\alpha \in \hat{H}^*(G, A), \beta \in \hat{H}^*(G, B), \gamma \in \hat{H}^*(G, C)$, we have:

$$(\alpha \smile \beta) \smile \gamma = \alpha \smile (\beta \smile \gamma)$$

2. If we identify $A \otimes B$ with $B \otimes A$, then for any $\alpha \in \hat{H}^p(G, A), \beta \in \hat{H}^q(G, B)$, we have

$$\alpha \smile \beta = (-1)^{pq} \beta \smile \alpha$$

3. If $H \leq G$ is a subgroup, then restriction is compatible with the cup product, i.e. for any $\alpha \in \hat{H}^*(G, A), \beta \in \hat{H}^*(G, B)$, we have

$$\text{res}_G^H(\alpha \smile \beta) = \text{res}_G^H(\alpha) \smile \text{res}_G^H(\beta)$$

4. If $H \triangleleft G$, then the inflation map is compatible with the cup product, i.e. for any $\alpha \in \hat{H}^*(G/H, A^H), \beta \in \hat{H}^*(G/H, B^H)$, we have

$$\inf_{G/H}(\alpha \smile \beta) = \inf_{G/H}(\alpha) \smile \inf_{G/H}(\beta)$$

5. If H is a subgroup of G , then the corestriction map is compatible with the cup product and the restriction map, i.e. for any $\alpha \in \hat{H}^*(H, A), \beta \in \hat{H}^*(G, B)$, we have

$$\text{cor}(\alpha \smile \text{res}(\beta)) = \text{cor}(\alpha) \smile \beta$$

Proof: The key idea is using dimension shifting to reduce to the case of $p = q = 0$.

We only prove 2. and 5.

For 2., if $p > 0$, consider the exact sequence of G -modules

$$0 \longrightarrow A \longrightarrow I(A) \longrightarrow A_1 \longrightarrow 0$$

Then we may take $\alpha = \delta\alpha_1$ and δ is an isomorphism.

Here we need to show that the above short exact sequence splits as a sequence of abelian groups to show that

$$0 \longrightarrow A \otimes B \longrightarrow I(A) \otimes B \longrightarrow A_1 \otimes B \longrightarrow 0$$

is exact.

Here we can identify $I(A)$ with $\mathbb{Z}[G] \otimes A$. Then

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}[G] \longrightarrow I_G \longrightarrow 0$$

splits as a sequence of abelian groups. Tensoring with A we get the desired splitting.

Use induction hypothesis we have $\alpha \smile \beta = \delta\alpha_1 \smile \beta = \delta(\alpha_1 \smile \beta) = \delta((-1)^{(p-1)q}\beta \smile \alpha_1) = (-1)^{(p-1)q}(-1)^q\beta \smile \delta\alpha_1 = (-1)^{pq}\beta \smile \alpha$.

If $p < 0$, consider the exact sequence of G -modules

$$0 \longrightarrow A_1 \longrightarrow I(A) \longrightarrow A \longrightarrow 0$$

and use the same argument as above.

Do the same for q and we reduce to the case of $p = q = 0$ and it is true by the construction of the cup product.

For 5., it is similar. We need to show that restriction and corestriction maps are compatible with the connecting homomorphisms.

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Remark: There are two important short exact sequences of G -modules: (which split as a sequence of abelian groups)

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}[G] & \longrightarrow & J_G \longrightarrow 0 \\ 0 & \longrightarrow & I_G & \longrightarrow & \mathbb{Z}[G] & \longrightarrow & \mathbb{Z} \longrightarrow 0 \end{array}$$

Example 6.9 : $G = \mathbb{Z}/m\mathbb{Z}$ with generator σ .

Then we have a exact sequence of G -modules:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\text{norm}} \mathbb{Z}[G] \xrightarrow{\sigma - 1} \mathbb{Z}[G] \xrightarrow{\text{aug}} \mathbb{Z} \longrightarrow 0$$

So we have $\Theta_{\mathbb{Z}} : \hat{H}^q(G, \mathbb{Z}) \cong \hat{H}^{q+2}(G, \mathbb{Z})$.

Similarly, tensoring above exact sequence (which splits as a sequence of abelian groups) with A , note that $\mathbb{Z}[G] \otimes A = I_G(A)$, which has vanishing Tate cohomology, we have $\Theta_A : \hat{H}^q(G, A) \cong \hat{H}^{q+2}(G, A)$ for any G -module A .

Now, we have $\hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/m\mathbb{Z}$. Using the isomorphism between $\hat{H}^0(G, \mathbb{Z})$ and $\hat{H}^2(G, \mathbb{Z})$, there is an element $a \in \hat{H}^2(G, \mathbb{Z})$ corresponding to $1 \in \mathbb{Z}/m\mathbb{Z}$.

We claim that Θ_A is given by cup product with a , i.e. $\Theta_A(\alpha) = \alpha \smile a$ for any $\alpha \in \hat{H}^q(G, A)$.

For any $\alpha \in \hat{H}^q(G, A)$, we have

$$\begin{aligned} \alpha \smile a &= \alpha \smile (\delta \circ \delta(1)) \\ &= (-1)^q \delta(\alpha \smile \delta(1)) \\ &= \delta(\delta(\alpha \smile 1)) \\ &= \delta \circ \delta(\alpha) \\ &= \Theta_A(\alpha) \end{aligned}$$

where the connecting homomorphisms are given by the short exact sequences given by above exact sequences of G -modules.

Lemma 6.10 : G : finite group. A, B : G -modules.

Let f be an inhomogeneous 1-cocycle of G with values in B .

Let $a \in A$ be an element with $N_G(a) = 0$.

Then f, a represent elements in $\hat{H}^1(G, B)$ and $\hat{H}^{-1}(G, A)$ respectively.

Then the cup product $a \smile f$ is represented by the inhomogeneous 0-cocycle

$$c = - \sum_{\sigma \in G} (\sigma \cdot a) \otimes f(\sigma)$$

Proof: Consider the exact sequence of G -modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}[G] \xrightarrow{\rho} J_G \longrightarrow 0$$

tensoring with B , since

$$\iota_* : \hat{H}^1(G, B) \rightarrow \hat{H}^1(G, \mathbb{Z}[G] \otimes B) = 0$$

we know that there is a $\beta \in \mathbb{Z}[G] \otimes B$ s.t. $f(\sigma) = \sigma\beta - \beta$.

Also, $\rho \circ f(\sigma) = 0$ for any $\sigma \in G$.

By the construction in snake lemma, we know that $f = \delta(\rho(\beta))$.

Then

$$\begin{aligned} a \smile f &= a \smile \delta(\rho(\beta)) \\ &= (-1)^{-1} \delta(a \smile \rho(\beta)) \end{aligned}$$

The connecting homomorphism here is given by the norm map between exact sequences of G -modules.

So we have

$$\begin{aligned} a \smile f &= -\delta(a \smile \rho(\beta)) \\ &= -N_G(a \otimes \beta) \\ &= -\sum_{\sigma \in G} (\sigma \cdot a) \otimes \sigma\beta \\ &= -\sum_{\sigma \in G} (\sigma \cdot a) \otimes f(\sigma) - \sum_{\sigma \in G} (\sigma \cdot a) \otimes \beta \\ &= -\sum_{\sigma \in G} (\sigma \cdot a) \otimes f(\sigma) \end{aligned}$$

Remark: The cup product with a 0-cocycle is given by tensor product. ■